

UCS301 Data Structures

Lab assignment 2 (Arrays)

1. Design a program to reverse the elements of an array without using extra space.
2. You are given an integer array `nums` containing $n + 1$ integers. Each integer is in the range $[1, n]$ (inclusive). Since there are $n + 1$ numbers but only n possible values, at least one number must appear more than once. Your task is to **find and return the duplicate number** without modifying the input array.
3. Design the logic to demonstrate transpose of a sparse matrix.
4. Design a program to demonstrate multiplication of two sparse matrix.
5. Given an $N \times M$ integer matrix, traverse its elements in a zigzag diagonal order and return the traversal as a one-dimensional array. The traversal follows these rules:
 - a. Start from the element at the top-left corner (0, 0).
 - b. Process the matrix one diagonal at a time, where each diagonal consists of elements having the same value of (row + column).
 - c. For even-numbered diagonals (row + column is even), traverse the diagonal from bottom to top.
 - d. For odd-numbered diagonals (row + column is odd), traverse the diagonal from top to bottom. Continue until every element of the matrix has been visited exactly once. Return the elements in the order they are visited.